

Skills Summary

Numerical, Graphical and Algebraic Ideas of a Limit

Use this reference while completing your worksheet or when studying.

Skill 1 — Limits by algebra

Rule: try substitution first. If it gives a number, that is the limit. If it gives $\frac{0}{0}$, that is not an answer — it is a signal that top and bottom share a factor. Factor and cancel (or multiply by the conjugate), then substitute into what is left. The cancelled function agrees with the original everywhere except at the point itself, and a limit never looks at the point itself.

Example: Find $\lim_{x \rightarrow 2} \frac{x^2 - 4}{x - 2}$.

Step 1. Substitution gives $0/0$, so the factor $x - 2$ must be common to top and bottom — factor and cancel it.

Step 2. $\frac{(x - 2)(x + 2)}{x - 2} = x + 2$ for $x \neq 2$, and substituting gives $2 + 2$.
 $\Rightarrow$ the limit is **4**

Try it: Find $\lim_{x \rightarrow 5} \frac{x^2 - 25}{x - 5}$.

check: 10

Skill 2 — Limits from a table of values

Rule: evaluate the function at inputs closing in on the target from both sides. If both columns settle on the same number, that number is the limit — but a table only *suggests* a limit. It shows finitely many inputs; the limit is a claim about all of them, so confirm it algebraically whenever you can.

Example: Evaluate $f(x) = \frac{x^2 - 4}{x - 2}$ at $x = 1.99$.

Step 1. The function is undefined at 2, so approach it: pick an input just below the target and read the output.

Step 2. $\frac{1.99^2 - 4}{1.99 - 2} = \frac{-0.0399}{-0.01}$.
 $\Rightarrow f(1.99) = \mathbf{3.99}$, already close to the value 4 that algebra confirms.

Try it: Evaluate the same f at $x = 2.01$.

check: 4.01

Skill 3 — One-sided limits from a graph

Rule: $\lim_{x \rightarrow a} f(x)$ exists only when the left-hand and right-hand limits agree. On a graph, trace the curve toward a from the left and from the right and read off the heights the curve heads for — not the plotted value at a , which may be a separate dot or missing entirely.

Example: A function equals $3x - 1$ for $x < 1$. Find its left-hand limit at $x = 1$.

Step 1. Approaching from the left uses only the piece defined for $x < 1$, so substitute into that piece.

Step 2. $3(1) - 1$.

$$\Rightarrow \lim_{x \rightarrow 1^-} f(x) = 2$$

Try it: The same function equals $x + 4$ for $x \geq 1$. Find its right-hand limit at $x = 1$.

check: $\frac{5}{2}$

Skill 4 — A series sum is a limit

Rule: an infinite sum is not an addition you ever finish. It is defined as $\lim_{N \rightarrow \infty} S_N$, the limit of the partial sums. For a geometric series with $|r| < 1$, $S_N = \frac{a(1 - r^N)}{1 - r}$, and since $r^N \rightarrow 0$ the limit is $\frac{a}{1 - r}$.

Example: Find $\sum_{n=1}^{\infty} \left(\frac{1}{2}\right)^{n-1}$.

Step 1. The ratio $1/2$ has absolute value under 1, so the partial sums converge and the limit formula applies.

Step 2. First term $a = 1$, ratio $r = \frac{1}{2}$, so the sum is $\frac{1}{1 - 1/2}$.

$$\Rightarrow \text{the sum is } 2$$

Try it: Find $\sum_{n=1}^{\infty} \left(\frac{1}{5}\right)^{n-1}$.

check: $\frac{5}{4}$